

Kaneko's Conjecture and its Extensions

Consider the group $SL_2(\mathbb{Z})$ of 2×2 matrices with integer entries and a determinant of 1. There is a group action of $SL_2(\mathbb{Z})$ on the upper-half plane of complex numbers defined by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az + b}{cz + d}.$$

A modular form f of weight k is a holomorphic function on the upper-half plane such that

$$f\left(\frac{az + b}{cz + d}\right) = (cz + d)^k f(z),$$

when $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, with additional boundedness conditions. An important weakly holomorphic modular form of weight 0 is Klein's modular invariant, denoted by $j(\tau)$. Though j is only defined on the upper-half plane, it is possible to consider the "values" of real quadratic irrationals through the following definition. Let w be a real quadratic irrational. Then it will be a fixed point of some $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = A \in SL_2(\mathbb{Z})$, and we can define the quadratic form $Q(x, y) = cx^2 + (d - a)xy - by^2$, up to a sign, with discriminant D . Then we define the value of w as

$$\text{val}_j(w) = \frac{1}{2 \ln(\varepsilon_w)} \int_{\tau}^{A\tau} \frac{j(z) \sqrt{D}}{Q(z, 1)} dz,$$

where τ is an arbitrary element of the upper half plane, ε_w is the largest eigenvalue of A . This project seeks to extend the conjectures of Kaneko [3] to modular forms of arbitrary weight. For the j -function, it has recently been proved by [2] that for every quadratic irrational w , $\text{Re}(\text{val}_j(w)) \geq \text{Re}(\text{val}_j(\phi))$, where ϕ denotes the golden ratio. For an arbitrary modular form f of weight $-2n$, we define the value of the quadratic irrational w to be

$$\text{val}_f(w) = \frac{1}{2 \ln(\varepsilon_w)} \int_{\tau}^{A\tau} f(z) \left(\frac{\sqrt{D}}{Q(z, 1)} \right)^{n+1} dz.$$

By the condition that f must have weight $-2n$, the above definition is well-defined independent of the choice of τ .

Computational Work

In [1], a formula is derived for the function val_j that allows it to be more easily calculated by a computer. In fact, the result can be easily extended to a general modular form of weight $-2n$. This form is also very useful in proving Kaneko's conjecture. Every real quadratic irrational has in its orbit under $SL_2(\mathbb{Z})$ a purely periodic continued fraction $[a_1, \dots, a_{2l}]$. Further, the value of a quadratic irrational is invariant under the action of $SL_2(\mathbb{Z})$, so we can choose this continued fraction expression for ease of calculation. The number $[a_1, \dots, a_{2l}]$ will be a fixed point of the matrix $T^{a_1} V^{a_2} \dots T^{a_{2l-1}} V^{a_{2l}}$, where

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

From the continued fraction representation of w , we can algorithmically calculate two sequences w_k and $\tilde{w}_k$ with $a_1 + \dots + a_{2l}$ entries such that the following identity holds:

$$\text{val}_f(w) = \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} f(e^{it}) i e^{-nit} \sum_{k=1}^{2l} \sum_{m=1}^{a_k} \left(\frac{e^{it}}{e^{it} - w_{k,m}} - \frac{e^{it}}{e^{it} - \tilde{w}_{k,m}} \right)^{n+1} dt$$

With this, we can start expression, we began to compute the values of various quadratic irrationals with the modular forms $\frac{E_4^2}{\Delta}$ and $\frac{E_4 E_6}{\Delta}$, of weights -4 and -2 respectively, where $E_{2k}(\tau) = \frac{1}{2\zeta(2k)} \sum'_{m,n} \frac{1}{(m+n\tau)^{2k}}$, where the summation symbol $\sum'$ denotes the sum over all pairs of non-negative integers (m, n) except for $(0, 0)$ and $\Delta = \frac{(2\pi)^{12}}{1728} (E_4^3 - E_6^2)$. A function E_{2k} is called an Eisenstein series.

Computational Results

First, we work with the modular form $f = \frac{E_4 E_6}{\Delta}$. We calculated numerically that $\text{Re}(\text{val}_f(\phi)) \approx -3.81 \cdot 10^{-14}$ and $\text{Re}(\text{val}_f([2, 2])) \approx -3.58 \cdot 10^{-14}$. However, for this specific case, it can actually be rigorously proven that $\text{Re}(\text{val}_f(\phi)) = 0$ and $\text{Re}(\text{val}_f([2, 2])) = 0$. The non-zero values are purely due to numerical errors. We also calculate

$$\text{Re}(\text{val}_f([1, 2])) \approx -26.26,$$

and

$$\text{Re}(\text{val}_f([4, 1])) \approx 34.72.$$

These results show that ϕ no longer has a minimum or maximum value for the case $f = \frac{E_4 E_6}{\Delta}$.

Now, we consider the case $f = \frac{E_4^2}{\Delta}$. We obtain the following table of values:

w	$\text{Re}(\text{val}_f(w))$
ϕ	99.93
$[1, 2]$	62.47
$[4, 2]$	5.494
$[1, 2, 3, 4]$	14.99
$[5, 1, 3, 2]$	12.18
$[100, 1]$	-19.94

This pattern continues for all the quadratic irrationals we checked. These results suggest that for the modular form $f = \frac{E_4^2}{\Delta}$, $\text{Re}(\text{val}_f(w))$ is in fact an upper bound on the values of quadratic irrationals.

A Useful Lemma

For a modular form f of weight $-2n$ with n even, the function $e^{-int} f(e^{it})$ will be real for $t \in [\frac{\pi}{3}, \frac{2\pi}{3}]$. This suggests that a useful way to write $\text{Re}(\text{val}_f(w))$ is

$$\int_{\pi/3}^{2\pi/3} e^{-int} f(e^{it}) \hat{S}_w^n(t) dt,$$

where

$$\hat{S}_w^n(t) = \frac{1}{2 \ln(\varepsilon_w)} \text{Re} \left(i \sum_{k=1}^{2l} \sum_{m=1}^{a_k} \left(\frac{e^{it}}{e^{it} - w_{k,m}} - \frac{e^{it}}{e^{it} - \tilde{w}_{k,m}} \right)^{n+1} \right).$$

When n is even, we can replace $\hat{S}_w^n(t)$ with function $S_w^n(t) = \frac{1}{2} (\hat{S}_w^n(t) + \hat{S}_w^n(\pi - t))$ without changing the value of the integral. This is useful, because $S_w^n(\pi - t) = S_w^n(t)$, so we can apply a lemma about to be introduced.

Now, we introduce the auxiliary function $D_{\phi, w}^n(t) = S_{\phi}^n(t) - S_w^n(t)$, so we can write

$$\text{Re}(\text{val}_f(\phi)) - \text{Re}(\text{val}_f(w)) = \int_{\pi/3}^{2\pi/3} e^{-nit} f(t) D_{\phi, w}^n(t) dt.$$

(Note: in the expression $D_{\phi, w}^n$, the n is acting a superscript and not an exponent.) If we can prove that this integral is always positive for any w , then it follows that ϕ has a maximal value for quadratic irrationals. We use the following lemma: Let $D : [\frac{\pi}{3}, \frac{2\pi}{3}] \rightarrow \mathbb{R}$ be a continuous function that satisfies $D(\pi - t) = D(t)$, is decreasing for $t \in [\frac{\pi}{3}, \frac{\pi}{2}]$, and has $\int_{\pi/3}^{2\pi/3} D(t) dt = 0$. Let $g : [\frac{\pi}{3}, \frac{2\pi}{3}] \rightarrow \mathbb{R}$ be a non-negative continuous function that satisfies $g(\pi - t) = g(t)$ and is increasing for $t \in [\frac{\pi}{3}, \frac{\pi}{2}]$. Then

$$\int_{\pi/3}^{2\pi/3} g(t) D(t) dt \leq 0.$$

In our case, we slightly change the lemma: if g and D are both non-positive and decreasing for $t \in [\frac{\pi}{3}, \frac{\pi}{2}]$, then the integral is greater than or equal to 0.

Extending the Lemma

We tried to fit this lemma to the case where $D(t) = D_{\phi, w}^2(t)$ and $g(t) = e^{-2it} \frac{E_4^2(e^{it})}{\Delta(e^{it})}$.

Except in the case $n = 0$, it is not in general true that $\int_{\pi/3}^{2\pi/3} D_{\phi, w}^n(t) dt = 0$.

It can be shown that $g(t) = g(\pi - t)$ for $t \in [\frac{\pi}{3}, \frac{2\pi}{3}]$. The following graph, calculated numerically, strongly suggests that $g(t)$ is strictly nonpositive on $[\frac{\pi}{3}, \frac{2\pi}{3}]$.

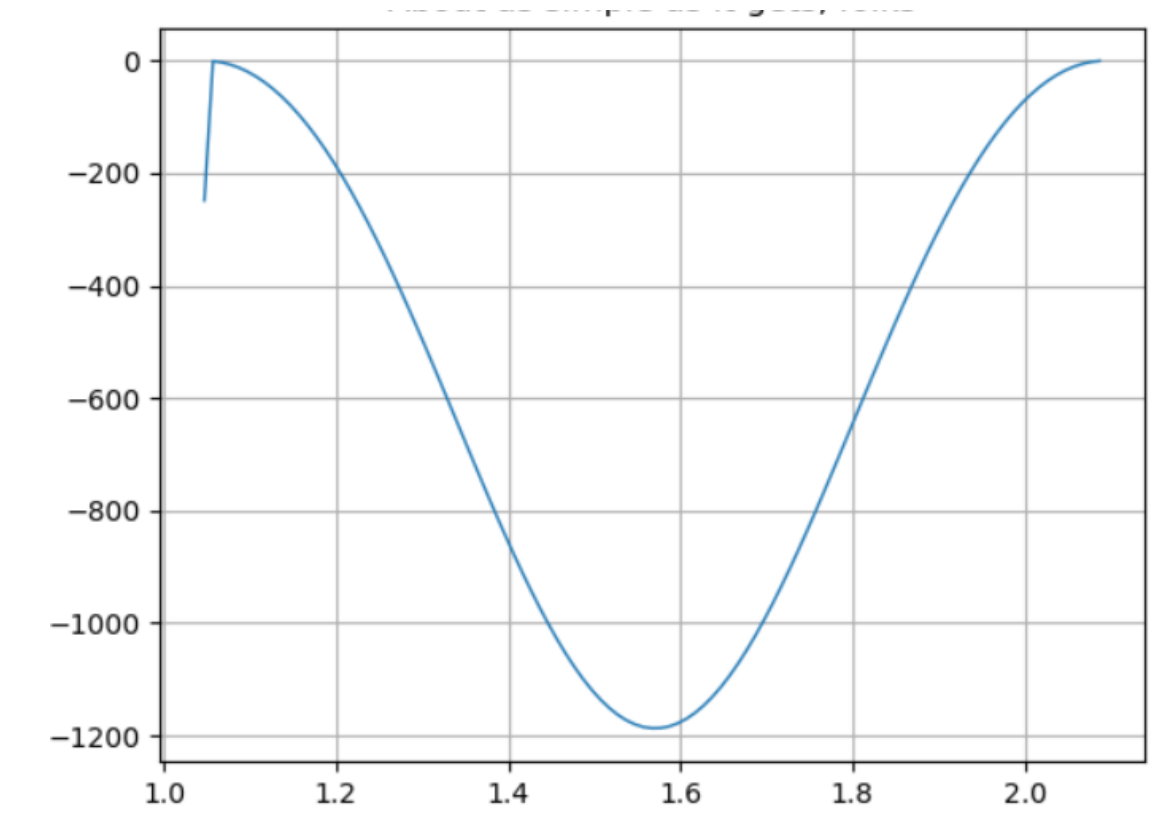


Fig. 1: A graph of $e^{-2it} f(e^{it})$ on the interval from $\frac{\pi}{3}$ to $\frac{2\pi}{3}$.

It can be computed (using Ramanujan's formulas for the derivatives of Eisenstein series) that

$$\frac{g'(t)}{g(t)} = -2i + \frac{2\pi}{3} e^{it} \left[E_2(e^{it}) + 2 \frac{E_6(e^{it})}{E_4(e^{it})} \right].$$

Thus, $g(t)$ satisfies all the required conditions if we can prove that

$$\text{Re} \left(e^{it} \left[E_2(e^{it}) + 2 \frac{E_6(e^{it})}{E_4(e^{it})} \right] \right) \geq 0$$

when $t \in [\frac{\pi}{3}, \frac{\pi}{2}]$.

Unfinished Work and Remaining Questions

- What further steps or techniques are required to complete the rigorous proof that $g(t)$ is decreasing on $[\frac{\pi}{3}, \frac{\pi}{2}]$?
- How can we refine the conjecture for the case where n is odd?
- Does this method work for other modular forms of weight $-2n$ for even n ?

References

- [1] P. Bengoechea, S. Herrero, and Ö. Imamoglu. *A Lyapunov Exponent Attached to Modular Functions*. 2025. arXiv: 2503.16343 [math.NT]. URL: <https://arxiv.org/pdf/2503.16343>.
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- [3] M. Kaneko. "Observations on the "Values" of the Elliptic Modular Function $j(\tau)$ at Real Quadratics". In: *Kyushu J. Math.* 63.2 (2009), pp. 353–364. URL: https://www.jstage.jst.go.jp/article/kyushujm/63/2/63_2_353/_article/-char/en.